25900273

**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

**MGU-UGP (HONOURS) Regular EXAMINATION October 2025**

**Third SEMESTER**

**Discipline Specific Elective (DSE) - MG3DSEBTG201,MG3DSEBBT202 -**

**PLANT AND ANIMAL PHYSIOLOGY**

**(2024 ADMISSION ONWARDS)**

Duration: 2 Hours Maximum Marks: 70

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word or sentence-Answer any 10 questions

Each question carries 2 marks

1. Name two functions of plasma protein. [K] / [CO1]
   Any two functions -2marks
2. Define reflex action. [K] / [CO1]
   Definition -2marks
3. What is diffusion? [K] / [CO3]
   Diffusion- definition- 2 marks
4. Name two microbes that increase organic matter in soil. [K] / [CO4]
   Any two microbes-2 marks
5. Name two compontents of circulatory system. [K] / [CO1]
   Two components- 2 marks
6. Which hormone can replace vernalization? [K] / [CO3]
   Gibberllin -2marks
7. What is the primary function of respiratory pigments ? [K] / [CO1]
   Transport of oxygen -2marks
8. Name an example of a natural cytokinin. [K] / [CO3]
   Zeatin- 2 marks
9. Name two examples of nitrogen - fixing bacteria. [K] / [CO4]
   Any two-2marks
10. Define macronutrients. [K] / [CO3]
    Definition, examples- 2 marks
11. Who is the" Father of Modern Physiology "and write the meaning of "Physiology". [K] / [CO1]

"Father of Modern Physiology"- Cluade Bernard- 1 mark, "Physiology". -Study of functions of living things -1mark.

12. Name the laboratory tests in the diagnosis of endocrine system disorders. [K] / [CO2]
    Any two tests -2marks

[2x10 = 20]

Part B

Short Essay-Answer any 4 questions

Each question carries 5 marks

13. Explain the deficiency symptoms of any five macro elements. [U] / [CO3]
    Deficiency symptoms of any five macro elements- 5 marks
14. Leaves of Mimosa plant are sensitive to touch. Explain. [U] / [CO3]
    Nastic movements- introduction- 2 marks, explain the mechanism-3 marks
15. Explain the traditional methods of crop improvement. [U] / [CO4]
    Conventional crop improvement methods-selection, hybridization etc. - explain- 5 marks
16. Explain digestion and why it is important for survival? [U] / [CO1]
    Digestion- definition- 1 mark, explanation – 2 marks, importance- 2 marks
17. Discuss on the symptoms of lack of oxygen to the brain. [U] / [CO2]
    Explain any five symptoms- 5 marks
18. Explain the different types of water absorption in plants. [U] / [CO3]
    Water absorption- introduction- 1 mark, types of water absorption-explain- 4 marks

[5x4 = 20]

Part C

Essay type questions-Answer any 2 questions

Each question carries 15 marks

19. Discuss biofertilizers types, benefits and examples of microbes used in agriculture. [U] / [CO4]
    Biofertilizers- definition- 3 marks, types -5 marks, benefits- 5 marks, eamples-2 marks
20. Describe the process which is common in aerobic and anaerobic respiration. [U] / [CO3]
    Aerobic and anaerobic respiration- introduction- 3 marks, explain the process-
21. Describe common nutritional disorders causes, symptoms and preventive measures. [U] / [CO1]
    Common nutritional disorders- 3 marks, explain the causes and symptoms-7 marks, preventive measures- 5 marks
22. Discuss on the common conditions, the disorders that affect lungs and its treatment?[U] / [CO2]
    Disorders that affect lungs- explain its causes and symptoms- 8 marks, Treatments- 7 marks

[15x2 = 30]